

Customer Shopping Behavior Analysis

Python • MySQL • Tableau Public

Executive Summary

This project analyses 3,900 customer transactions to identify spending patterns, customer segments, product preferences, discount behaviour, shipping choices, and subscription trends. Python was used for data preparation, MySQL for business analysis, and Tableau Public for dashboard visualisation.

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1. Dataset Overview

Attribute	Details
Records	3,900 customer purchases
Columns	18 variables
Demographics	Age, gender, location, and subscription status
Purchase details	Item, category, purchase amount, season, size, and colour
Shopping behaviour	Discount use, previous purchases, purchase frequency, review rating, shipping type, and payment method
Missing values	37 missing values in Review Rating

2. Analytical Workflow

1. Load and inspect the source dataset in pandas.
2. Clean missing values and standardise column names.
3. Engineer analytical fields such as age_group and purchase_frequency_days.
4. Validate redundant fields and remove promo_code_used after confirming overlap with discount_applied.
5. Load the cleaned DataFrame into MySQL.
6. Run SQL queries to answer business questions.
7. Present the results in an interactive Tableau Public dashboard.

3. Data Preparation in Python

- **Data loading:** Imported the dataset using pandas.
- **Initial exploration:** Used df.info() to inspect structure and df.describe() to review summary statistics.
- **Missing-value treatment:** Checked null counts with df.isnull().sum() and imputed Review Rating values using the median rating within each product category.
- **Column standardisation:** Converted column names to snake_case for consistent querying and documentation.

- **Feature engineering:** Created age_group by binning customer ages and purchase_frequency_days from purchase-frequency data.
- **Consistency validation:** Confirmed that discount_applied and promo_code_used were redundant, then removed promo_code_used.
- **Database integration:** Connected Python to MySQL and loaded the cleaned DataFrame for SQL analysis.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Figure 1. Summary statistics generated during initial dataset exploration.

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

Figure 2. Continuation of the summary output for shopping-behaviour fields.

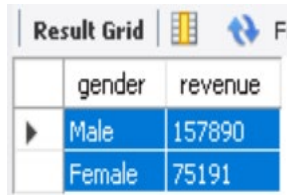
4. Business Analysis in MySQL

Structured SQL queries were used to answer ten business questions and translate transaction data into decision-ready findings.

4.1 Revenue by Gender

Total purchase revenue was compared across male and female customers.

Key finding: Male customers generated 157,890 in revenue, compared with 75,191 from female customers.



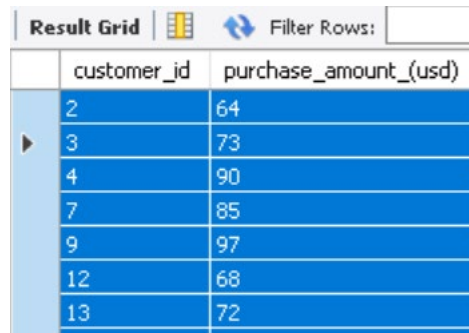
	gender	revenue
▶	Male	157890
	Female	75191

Figure 3. Revenue by gender query result.

4.2 High-Spending Discount Users

Customers who received a discount and still spent above the overall average purchase amount were identified.

Key finding: Discount usage did not necessarily correspond to low-value purchases; several discounted transactions remained above average.



	customer_id	purchase_amount_(usd)
▶	2	64
	3	73
	4	90
	7	85
	9	97
	12	68
	13	72

Figure 4. Sample of high-spending customers who used discounts.

4.3 Top Five Products by Rating

Products were ranked by average review rating.

Key finding: Gloves ranked first at 3.86, followed by Sandals (3.84), Boots (3.82), Hat (3.80), and Skirt (3.79).



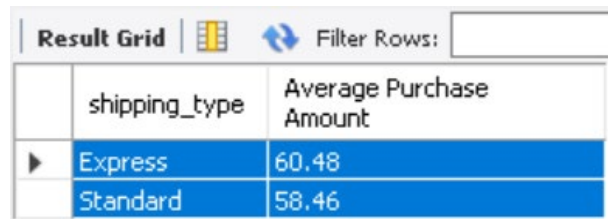
	item_purchased	AVG. PRODUCT RATING
▶	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Skirt	3.79

Figure 5. Highest-rated products by average review score.

4.4 Shipping Type Comparison

Average purchase amounts were compared for Standard and Express shipping.

Key finding: Express shipping produced a slightly higher average purchase amount (60.48) than Standard shipping (58.46).



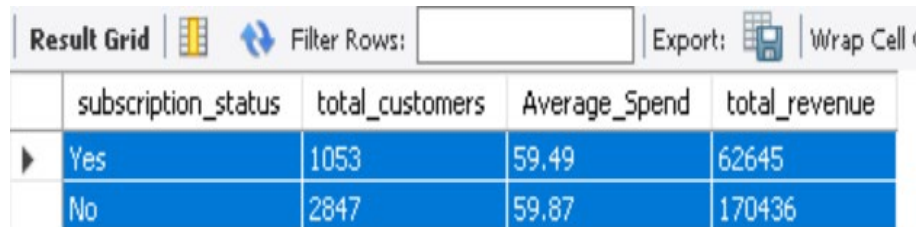
	shipping_type	Average Purchase Amount
▶	Express	60.48
	Standard	58.46

Figure 6. Average purchase amount by selected shipping type.

4.5 Subscribers vs. Non-Subscribers

Customer count, average spend, and total revenue were compared by subscription status.

Key finding: Average spend was similar for subscribers (59.49) and non-subscribers (59.87), while non-subscribers generated more total revenue because they represented the larger customer group.



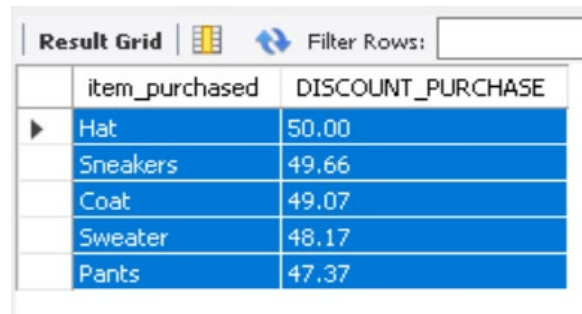
	subscription_status	total_customers	Average_Spend	total_revenue
▶	Yes	1053	59.49	62645
	No	2847	59.87	170436

Figure 7. Subscription status, customer count, average spend, and revenue.

4.6 Discount-Dependent Products

Products were ranked by the percentage of purchases made with a discount.

Key finding: Hat had the highest discounted-purchase share at 50.00%, followed by Sneakers, Coat, Sweater, and Pants.



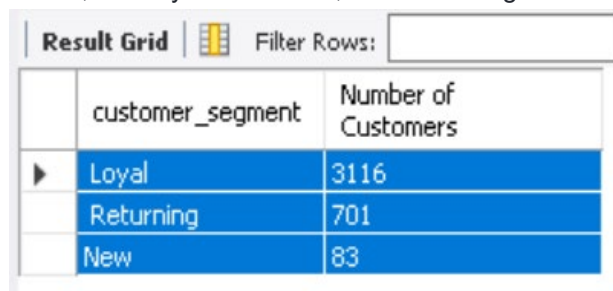
	item_purchased	DISCOUNT_PURCHASE
▶	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

Figure 8. Products with the highest percentage of discounted purchases.

4.7 Customer Segmentation

Customers were classified as New, Returning, or Loyal based on previous purchase history.

Key finding: The dataset contains 3,116 Loyal customers, 701 Returning customers, and 83 New customers.



	customer_segment	Number of Customers
▶	Loyal	3116
	Returning	701
	New	83

Figure 9. Customer counts by behavioural segment.

4.8 Top Products within Each Category

Products were ranked within category using purchase-frequency measures.

Key finding: Scarf, Blouse, Shoes, and Coat appear as the leading products in their displayed categories.

Result Grid	Filter Rows:	Export:	Wrap Cell Co
item_purchased	category	avg_purchase_frequency	item_rank
Scarf	Accessories	100.2357	1
Backpack	Accessories	91.4895	2
Belt	Accessories	90.7826	3
Blouse	Clothing	105.2807	1
Socks	Clothing	103.4403	2
Dress	Clothing	101.7831	3
Shoes	Footwear	97.4867	1
Sandals	Footwear	93.2750	2
Sneakers	Footwear	84.3172	3
Coat	Outerwear	86.2422	1
Jacket	Outerwear	83.7853	2

Figure 10. Leading products within Accessories, Clothing, Footwear, and Outerwear.

4.9 Repeat Buyers and Subscription Status

The analysis reviewed subscription status among customers with more than five previous purchases.

Key finding: In absolute terms, 2,583 repeat buyers were non-subscribers and 980 were subscribers. A rate-based comparison is required to establish subscription propensity.

Result Grid	Filter Rows:
subscription_status	repeat buyers
Yes	980
No	2583

Figure 11. Repeat-buyer counts by subscription status.

4.10 Revenue by Age Group

Total revenue was aggregated across the engineered age groups.

Key finding: Young Adults generated the highest revenue at 62,143, followed by Middle-Aged customers at 59,197.

Result Grid	Filter Rows:
age_group	Total_revenue
Young Adult	62143
Middle-Aged	59197
Adult	55978
Senior	55763

Figure 12. Revenue contribution by age group.

5. Interactive Dashboard

An interactive dashboard was developed in Tableau Public to consolidate the principal insights. It includes customer and average-purchase KPIs, subscription distribution, revenue by category, revenue by age group, shipping-type comparisons, and interactive filters.

CUSTOMER BEHAVIOUR DASHBOARD

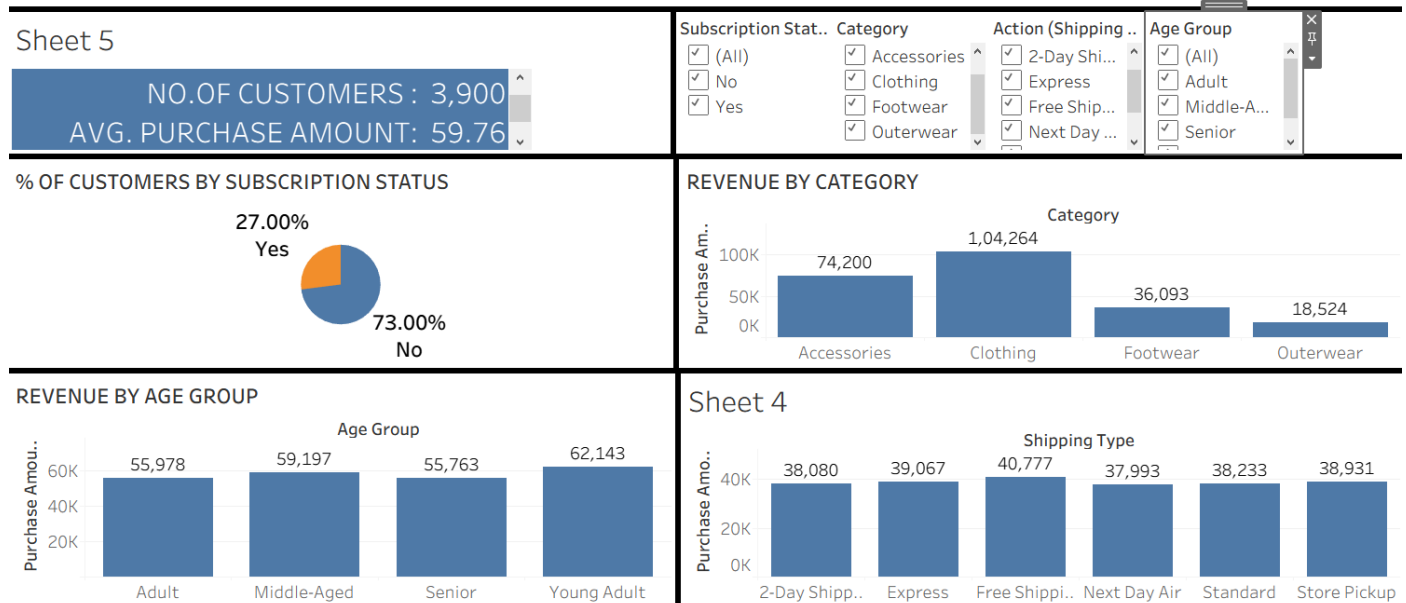


Figure 13. Customer Behaviour Dashboard created in Tableau Public.

6. Business Recommendations

- **Increase subscription conversion:** Promote clear subscriber-only benefits and target high-value non-subscribers.
- **Strengthen loyalty programmes:** Reward repeat buyers and create incentives that move Returning customers into the Loyal segment.
- **Optimise discount strategy:** Review products with high discount dependence to protect margins while retaining conversion benefits.
- **Prioritise high-performing products:** Feature top-rated and category-leading products in campaigns, recommendations, and merchandising.
- **Use targeted marketing:** Focus campaigns on high-revenue age groups and customers selecting higher-value shipping options.

7. Conclusion

The analysis demonstrates an end-to-end workflow from data cleaning and feature engineering to SQL-based business analysis and dashboard communication. The findings provide a practical foundation for improving subscription conversion, customer retention, product positioning, and promotional efficiency.

Tools: Python (pandas) • MySQL • Tableau Public